Jonathan Parlett

Curriculum Vitae

jp86440@uga.edu
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Department of Mathematics
University of Georgia

Education

2025– **University of Georgia**
Ph.D. Mathematics, in progress

2020–2025 **Drexel University**
B.S. Mathematics; B.S. Computer Science
GPA 3.8

Research Interests

Arithmetic geometry, algebra, and combinatorics.

Publications

2025 Top to random and reverse: analysis of a new descent algebra shuffle, joint with Darij Grinberg. Preprint. [arXiv]

2025 Fixed Point Homing Shuffles. *Discrete Mathematics & Theoretical Computer Science*, vol. 27:1. [DOI]

2023 Subnet Communicability: Diffusive Communication Across the Brain Through a Backbone Subnetwork, joint with Abhishek Jeyapratap, Ali Shokoufandeh, Birkan Tunc, and Yusuf Osmanlioglu. *CDMRI 2023*. [Springer]

Talks and Posters

2025 Long Paths in Polynomial Divisor Graphs, joint with Nicole Froitzheim. Poster, Joint Mathematics Meetings. [poster]

2024 Long Paths in Polynomial Divisor Graphs, joint with Kayla Traxler. YMC 2024, Ohio State University. [slides]

2023 Subnet Communicability, joint with Abhishek Jeyapratap. CDMRI 2023. [slides]

2023 Subnet Communicability. Drexel Week of Undergraduate Excellence. [video]

Teaching

University of Georgia

Fall 2026 Instructor of record, Pre-calculus (Math 1113)

Fall 2026 Grader, Graduate Algebra I

Spring 2026 Grader, Introduction to Higher Mathematics

Fall 2025 Grader, Applied Linear Algebra

2020–2025 **Drexel University**
Tutor and grader in the computer science and mathematics departments, supporting data structures and algorithms, programming languages, algorithmic number theory and cryptography, and abstract and linear algebra. Responsibilities included holding office hours, grading, and occasionally creating supplemental course material.

Professional Experience

2022 **SRI International**
Intern, Data Analysis and Software Engineering Princeton, NJ March–September

- Worked under Principal Research Engineer Norman Hurst to develop a technology allowing recovery of time from video streams, robust against framerate conversions.
- Conducted statistical analysis of system performance to facilitate improvement and validation of functionality, and developed methods to further improve its accuracy. Contributions led to inclusion as co-inventor on a patent.
- Worked primarily in C, with performance analysis through Python and pandas.

2015–2020 **United States Marine Corps**
Data Systems Administrator Camp Pendleton, CA October 2015–February 2020

- Configured and maintained information systems networks to facilitate communication between military units in remote locales.
- Provided services such as voice, video, email, file sharing and internet, through a combination of Cisco and Microsoft software and hardware solutions.
- Trained peers and subordinates in occupational skills.

Awards

Spring 2025 Robert J. Bickel Endowed Scholarship

Spring 2024 Outstanding Undergraduate Teaching Assistant Award

Spring 2024 Yilin Yang Outstanding Undergraduate Math Research Award

Winter 2023 Undergraduate Research Mini-Grant

Dean's List, Drexel University

2018, 2020 Navy and Marine Corps Achievement Medal, United States Marine Corps